Wasel
COD Order Risk Assessment

● High 87 → ● Low 16

Check the risky orders before the courier leaves.

Wasel reads cash-on-delivery orders from chat, scores the chance the customer refuses, and drafts the one question that lowers it.

A refused parcel is a wasted trip.

One vendor guide puts refusals at **15 to 25 %** for Tunisian fashion orders. The seller pays for the courier either way.

Customer orders in chat

Darija, French, Arabic

Seller books a courier

The parcel is packed and paid for

Courier knocks

The address was "near the mosque"

Customer says no

The seller pays for the trip

Industry-reported figure, Kolonell 2026. No independent national number exists.

Customers write the way they talk.

ARABIZI

Douar Ouled Ali, 9rib el jemaa

Douar Ouled Ali, near the mosque

ARABIC SCRIPT

في نابل، قريب الجامع

FRENCH

je veux commander le jean slim. rue de Marseille, à côté de la poste, Sousse

MIXED, IN ONE MESSAGE

ama ghali chwaya. mouch dima fi dar

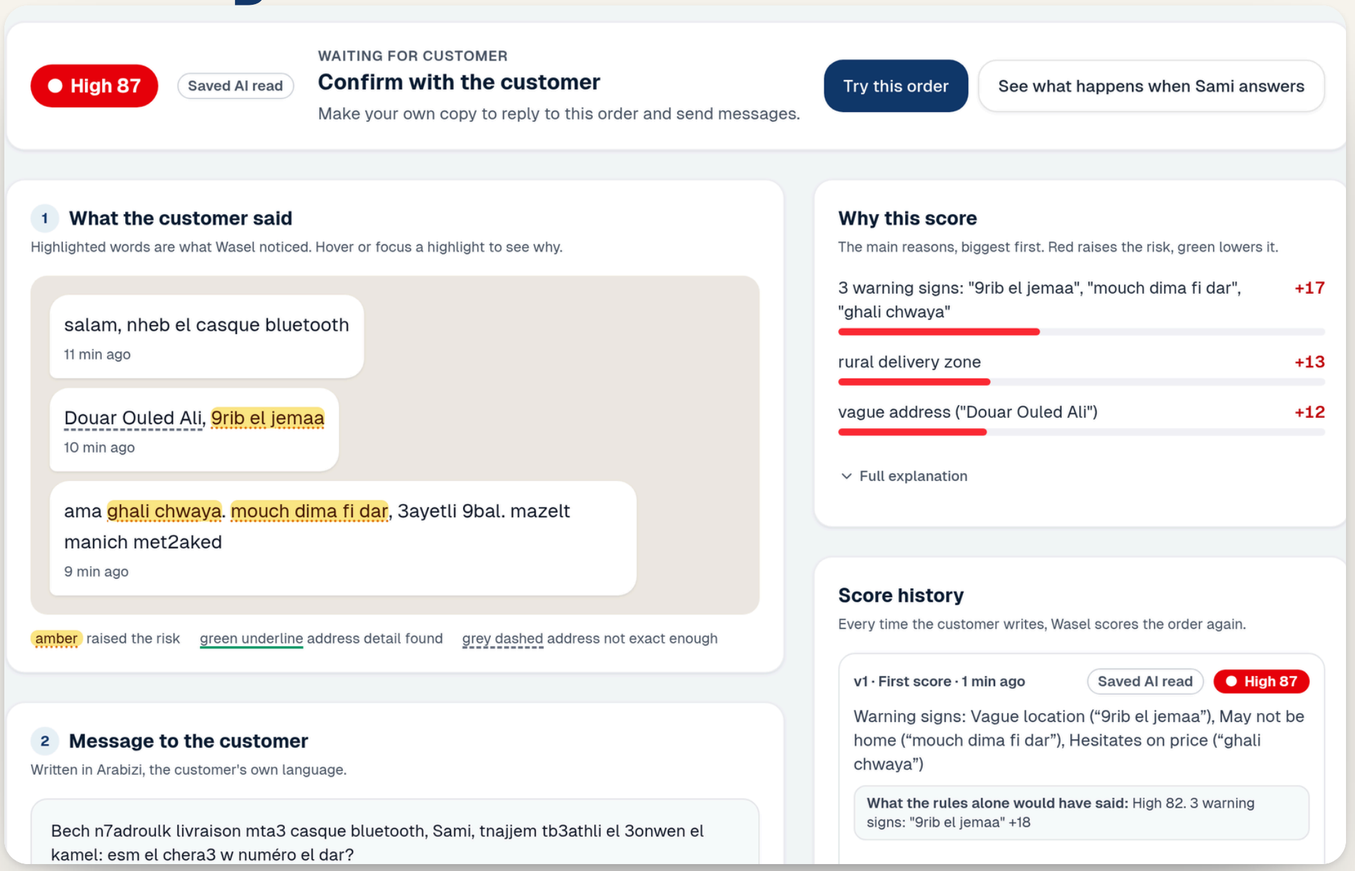
but it's a bit expensive. I'm not always home

Sellers read these by eye, one at a time. Wasel reads all of them and quotes the words that worry it.

Messages from the demo data. Highlight marks the words Wasel flags.

Wasel scores the order and shows why.

1. The customer's own words are highlighted.
2. Each reason carries its points.
3. The seller decides. Wasel drafts the question.



Order #2601, public demo, synthetic data.

One question takes it from 87 to 16.

Before

salam, nheb el casque bluetooth

Douar Ouled Ali, 9rib el jemaa

ama ghali chwaya. mouch dima fi dar, 3ayetli 9bal

● High 87

3 warning signs	+17
rural delivery zone	+13
vague address	+12

Wasel asks one question

Bech n7adroulk livraison mta3 casque bluetooth, Sami, tnajjem tb3athli el 3onwen el kamel: esm el chera3 w numéro el dar?

To prepare your delivery, could you send the full address: street name and house number?

→ **The customer answers**

ey behi. Douar Ouled Ali, triq el jemaa, dar 14, el bab lazra9 9odem el pharmacie. ghodwa ba3d 17h nkoun fi dar

OK. Street, house 14, blue door opposite the pharmacy. Tomorrow after 5 pm I'm home

Rescored

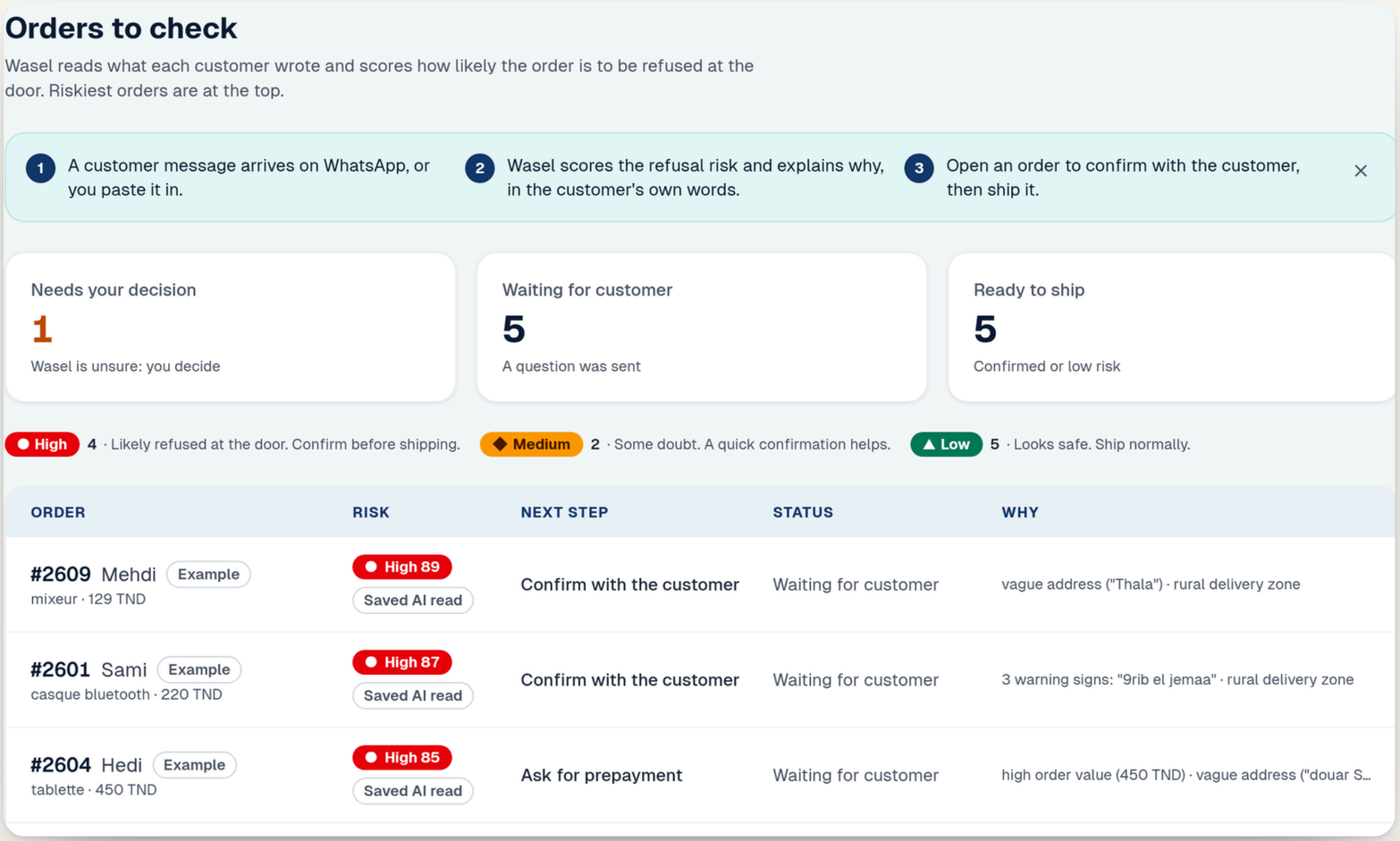
● Low 16

71 points lower. Ship it.

full address given	−20
rural delivery zone	+11
→ high order value (220 TND)	+9

Real order #2601 from the live demo. English lines are our own glosses.

The riskiest orders come first.



Each row ends in one next step.

High Confirm, or ask for prepayment

Medium One quick question

Low Ship it

A person sends every message unless the seller switches on automatic sending for clear-cut replies.

Order queue, public demo, synthetic data.

The model reads. Code scores.



Extract

Fills a 12-field schema from the chat. Rules take over if no model answers.

Verify

Every quote must appear in the original text, or it is dropped.

Score

XGBoost with monotone limits, plus rules in `scoring.yaml`.

Draft

Approved templates in the customer's language, checked three ways.

Send

A person approves what goes out.

The language model never picks a number. That is how we can show the reason behind every score.

A customer who writes "ignore previous instructions" is flagged. The order is still scored, and the address stays unconfirmed.

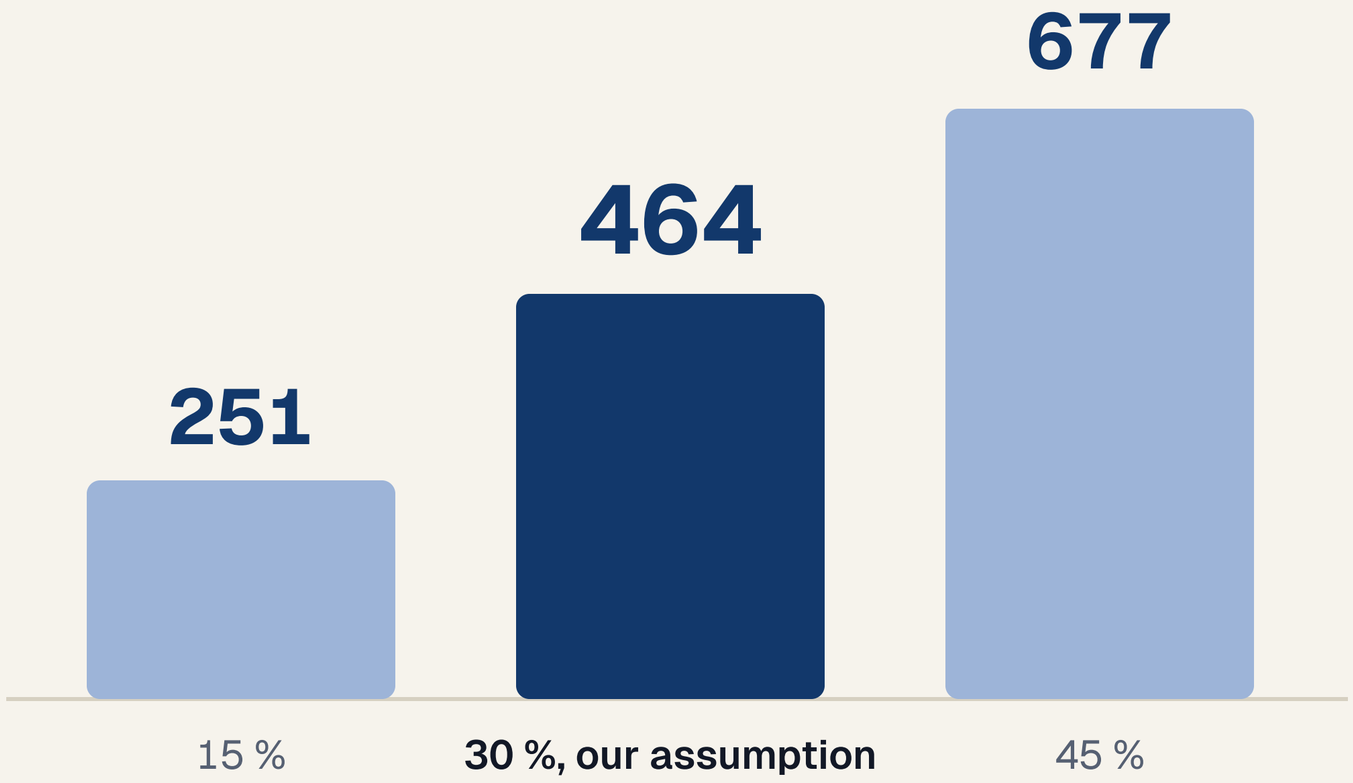
Tested on every change.

Regression gate: 279 extraction, 165 outbound and 32 injection cases

96.1 %	of orders recognised
97.1 %	of addresses within one level
0	quotes missing from the source text
165 / 165	unsafe drafts blocked or held
32 / 32	injection attempts flagged

Offline run, no keys. 737 tests. Full report: docs/evaluation.md

A simulated 464 TND saved per 1,000 orders.



Share of asked orders where a confirmation question prevents the refusal

TND saved per 1,000 orders, by how often a confirmation question prevents a refusal.

Refusals fall from 30.0 % to 23.6 % on 2,001 held-out orders in the simulation.

A failed trip costs 8 TND. A prepayment request prevents 60 % of refusals and loses 10 % of good orders.

The refusal labels are simulated, not observed. A pilot with a real seller replaces them.

data/reports/impact.json. The app labels this figure "simulated" wherever it appears.

AMD Token Factory comes first in the route.

1	**AMD Token Factory** Tried first when its key, address and model are set	Key pending
2	**Groq, gpt-oss-120b** Serves the live demo today	Running
3	**Gemini** Off unless the seller enables it	Optional
4	**Rules extractor** Needs no provider and no key	Always on

The route is built and covered by tests. Wasel keeps working with any provider missing, down to the rules extractor.

When the key arrives: `make tf-check`, then `make eval-live`.

What is built, and what comes next.

Built

- The pipeline from chat to score, draft and rescore
- The signed WhatsApp webhook path
- An offline gate that runs on every change
- A live public demo you can open now

Next

- A pilot with one real seller
- Refusal labels from real orders
- Native review of every template
- Token Factory evaluation on AMD

Three things to remember.

1. The seller sees the refusal risk before the courier leaves.
2. The model reads the chat. Code checks every quote and sets the score.
3. Wasel keeps working when any one provider is missing.

Live demo wasel.runs-on.dev · open order #2601

Code github.com/omarbounawarapy/wasel

